Lab 4 Written Questions

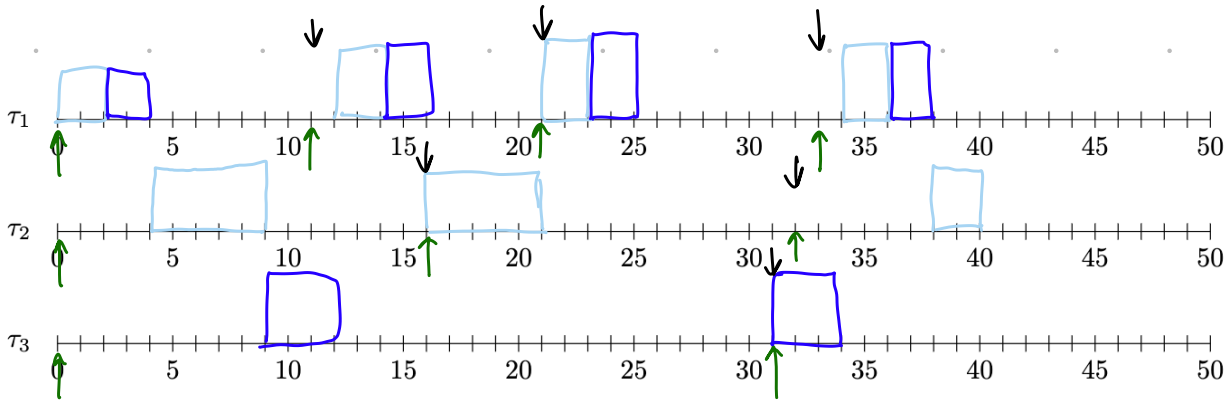
(1) $\tau_1 = (4, 11)$, $\tau_2 = (5, 16)$, $\tau_3 = (3, 31)$

$U = \frac{4}{11} + \frac{5}{16} + \frac{3}{31} = 0.3636 + 0.3125 + 0.0968 = 0.7729$

$UB_3 = 3(2^{1/3} - 1) = 0.7798$

$0.7798 \leq 0.7798 \rightarrow$ schedulable under RMS ✓

no task misses deadline



holding m1 running deadline release

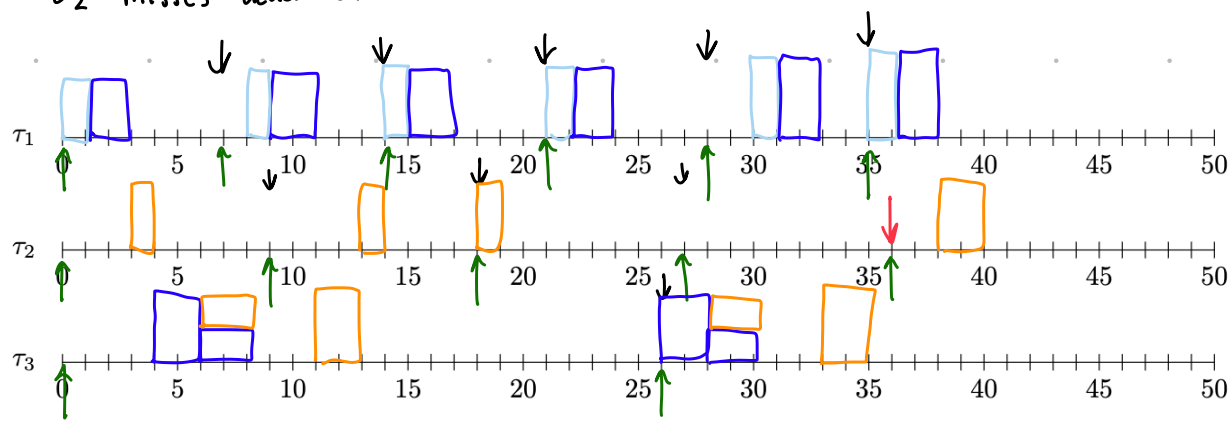
(2) $\tau_1 = (3, 7)$, $\tau_2 = (1, 9)$, $\tau_3 = (6, 26)$

$\frac{3}{7} + \frac{1}{9} + \frac{6}{26} = 0.4286 + 0.1111 + 0.2308 = 0.7705$

$n(2^{1/n} - 1)\Big|_{n=3} = 0.7798$

$0.7705 \leq 0.7798 \rightarrow$ task is schedulable! ✓

τ_2 misses deadline!



holding m1 running deadline release holding m2 missed deadline